

CMPSC 445 (Spring 2026)

Machine Learning

Project 1

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Project Description

The purpose of this project is to determine the primary causes of global temperature changes through the application of a regression-based data analysis process. A variety of environmental data sets were collected, cleaned, and combined to create a single data set that includes temperature anomaly data as well as atmospheric and emission-related data. The three data sources I used for the project are:

- NASA global temperature anomaly dataset (monthly)
- NOAA Mauna Loa CO₂ concentration dataset (monthly)
- Our World in Data (OWID) global emissions dataset (yearly)

The datasets were preprocessed and aligned by date, with yearly emissions being added to the monthly dataset using a common year attribute. Feature engineering techniques were used to create new features such as rolling averages (12-month moving averages), percentage growth rates, and a time index to reflect long-term trends.

Data Preprocessing

The preprocessing phase involved multiple steps to ensure data consistency and usability. Any missing values ('***' in NASA data) were converted to Nan and removed using dropna. Monthly temperature data was reshaped into long format. For NOAA CO₂, the data was cleaned by removing invalid values like -99.99. OWID data was filtered to include only global ("World") records. The datasets were merged using date (monthly) and year (yearly bridge).

Table 1 shows sample rows from the beginning of the datasets.

	date	temp	co2_noaa	...	co2_ma12	temp_ma12	days_since_start
11	1965-12-01	-0.08	319.42	...	320.036667	-0.106667	334
12	1966-01-01	-0.19	320.62	...	320.135000	-0.115833	365
13	1966-02-01	-0.04	321.60	...	320.231667	-0.105000	396

Table 2 shows more sample rows from a later time to demonstrate consistency.

	date	temp	co2_noaa	...	co2_ma12	temp_ma12	days_since_start
305	1990-06-01	0.37	356.39	...	353.802500	0.406667	9282
267	1987-04-01	0.24	351.40	...	348.148333	0.180000	8125
624	2017-01-01	1.02	406.36	...	404.715000	0.995000	18993

Modeling and Evaluation

The project required the dataset to be split 70-30. Seventy percent go into training and thirty percent go into testing sets.

Linear Regression Results

- R^2 Score: 0.8909
- Mean Squared Error (MSE): 0.01398

Random Forest Results

- R^2 Score: 0.9059
- Mean Squared Error (MSE): 0.01206

The Random Forest model slightly outperformed Linear Regression, indicating that nonlinear relationships exist between environmental variables and temperature. However, both models achieved high R^2 values (above 0.89), demonstrating strong predictive capability.

===== MODEL PERFORMANCE =====

Linear Regression:

R2: 0.8909165121254341

MSE: 0.013976580292706977

Random Forest:

R2: 0.905891698560926

MSE: 0.012057849055816366

===== FEATURE IMPORTANCE =====

Top Features (Linear Regression):

temp_ma12	0.310645
methane	0.217447
co2_ma12	-0.201401
co2	0.127051
energy_per_capita	-0.102409
nitrous_oxide	-0.042191
co2_noaa	0.032866
co2_growth	0.017336
methane_growth	-0.013164
days_since_start	-0.004822

dtype: float64

Top Features (Random Forest):

temp_ma12	0.608137
co2_ma12	0.144892
days_since_start	0.122755
co2_noaa	0.036496
co2	0.031308
co2_growth	0.022025
co2_growth_prct	0.010577
nitrous_oxide	0.008884
energy_per_capita	0.006057
methane	0.004704

dtype: float64

Driving Factors of Temperature Change

Based on the model results, the variables related to CO₂, such as co2_ma12, co2_noaa, and co2, are included in the top predictors in both Linear Regression and Random Forest models. In

particular, the high importance of the variable `co2_ma12` indicates that long-term trends of atmospheric CO₂ concentrations have a major influence on temperature change.

Human vs Natural Factors

The results strongly support the conclusion that human-driven factors dominate natural factors in influencing global temperature change.

- Human-driven factors:
 - CO₂ emissions (strongest influence)
 - Methane and other greenhouse gases
 - Energy consumption (indirectly reflected through emissions data)
- Natural factors:
 - Not directly included in the dataset
 - No evidence from the model suggesting comparable influence

Because the models achieve high predictive accuracy ($R^2 \approx 0.90$) using only human-related variables, it can be inferred that human activities are the dominant contributors to recent global warming trends, while natural factors play a smaller or secondary role in this analysis.

Discussion and Conclusion

The strongest variables contributing to global temperature change are related to CO₂ variables, with `co2_ma12` and `co2_noaa` consistently ranked as two of the most important variables across all models. This suggests a strong impact of long-term accumulation of CO₂ on temperature trends. Methane variables are also seen to impact temperature variation, although to a lesser

extent than CO₂ variables. Although temp_ma12 is ranked as the most important variable, this variable actually reflects temporal dependence rather than environmental cause. This result agrees with the existing climate science that states greenhouse gases, especially CO₂, are the primary cause of global warming. The importance of averages also lends credence to the notion that climate change is the result of cumulative, rather than fluctuating, environmental changes.

Some challenges were experienced during the process, including the format of the data, the existence of missing or invalid data, and the complexity of combining data with differing scales. Another challenge with the model is that it only shows correlation and not causation, and some natural influences, such as solar and volcanic activity, were not included in the data set.

From an ethical perspective, it is important to guarantee the reliability, transparency, and interpretation of the data, particularly when dealing with environmental data that has practical applications.

In conclusion, it is quite clear from the results obtained that greenhouse gases resulting from human activities are the key contributors to global temperature changes, with CO₂ being the most prominent of all. Although these models offer strong supporting evidence for these conclusions, it should be noted and considered in relation to other scientific research.

AI Usage

The use of AI tools was helpful in debugging, structuring the data pipeline, and improving code organization. Nevertheless, the overall project design, selection of data, data preprocessing logic, feature engineering, and interpretation were all done without any outside help. The integration of various datasets, management of missing values, and creation of regression models were all based on my understanding of data analysis and machine learning concepts.